



MULTIMEDIA UNIVERSITY OF KENYA

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

UNIVERSITY EXAMINATIONS 2016/2017

SECOND YEAR SECOND SEMESTER EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

BIT 2214: OBJECT ORIENTED ANALYSIS AND DESIGN

DATE: FRIDAY 7TH JULY 2017

TIME: 2 HOURS

INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED.

ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS.

QUESTION ONE (THIRTY MARKS)

(I) Explain the meaning of the term modeling (2marks)

(II) Outline five benefits of object oriented systems system analysis over the tradition systems analysis methodologies. (5marks)

Compare Object Oriented Analysis & Design & Structured System Analysis & Design. (4marks)

Explain the importance of UML as a modeling language (3marks)

In a Use Case Driven Object Oriented Analysis, Classes, objects Relationships, Attributes and methods are common terminology used. Explain with the aid of diagram the meaning of each of these terminologies. (6 marks)

Study the scenario below and create class diagrams for customer and order. Show all the possible attributes and operations.

We have customers who order our products. We distinguish corporate customers from personal customers, since corporate customers are billed monthly whereas personal customers need to prepay their orders with a credit card. (6 marks)

A deployment diagram is made up of 4 components. List and explain each one of these components. (4marks)

QUESTION TWO (TWENTY MARKS)

- a) (i) What do you mean by object-oriented design model? (2marks)
(ii) Discuss the input and output of object-oriented design. (4marks)
- b). Discuss the 4+1 view architecture of UML (6marks)
- c). Explain what you mean by class diagram. (2marks)
- d) Discuss the steps to draw the class diagram with any one example. (6marks)

QUESTION THREE (TWENTY MARKS)

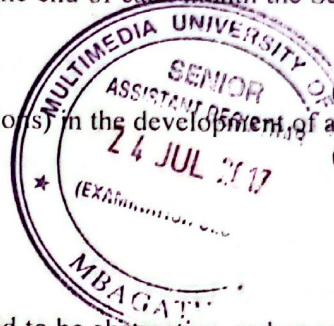
- a). Discuss object-oriented testing strategies in detail. (6marks)
- b) Differentiate between sequence diagram and collaboration diagram. (4marks)
- c). Draw the sequence diagram of login page. (4marks)
- d) Consider the application which is assumed to be a web based application which is deployed in a clustered environment using server 1, server 2 and server 3. The user is connecting to the application using internet. The control is flowing from the caching server to the clustered environment. So draw the deployment diagram of order management system. (6marks)

QUESTION FOUR (TWENTY MARKS)

- a) (I) Differentiate between system analysis and system design. (4marks)
(II) Explain how poor systems analysis and design may affect system development outcome. (6marks)
- b) *ABC Kids Playgroup* is an after school club for children under 12. The club wishes to keep information on the children and the staff who work there. Two types of staff are employed: Secretaries and Play Workers. Personal details, such as name and address are recorded for all staff and if they are a Play Worker they must also pass a police check. The system must record when this has been passed and when it must be renewed, because a Play Worker cannot work with the children until this condition is met. When a child starts at the club, a Secretary records personal details such as name, address, date of birth and at least one emergency contact number, up to a maximum of three. As part of the registration process, the Secretary also records who is allowed to collect the child and their contact details. The Play Worker will use these contact details, so that they can check that the person collecting the child at the end of a session is authorized to

make the collection. Some children have special circumstances (e.g. suffers from asthma) which are treated with medication, and the Secretary will record what the condition is, what medicine can be used and what to do in an emergency. There are different groups a child can join, after an initial assessment the Play Worker is responsible for allocating each child to a group appropriate to their age and ability. Each week the Play Worker will generate a letter for the parents to advise them which activities their children will take part in. At the end of each month the Secretary will generate an invoice for the parents to pay.

- (i) Draw a Use Case diagram for this system. (5marks)
(ii) Discuss the role of Use Cases (diagrams and descriptions) in the development of an object-oriented system. (5marks)



QUESTION FIVE (TWENTY MARKS)

- a) Four basic features of object-oriented paradigm are said to be abstraction, polymorphism, encapsulation and inheritance. Define each of these terms. (6marks)
- b) Describe the meaning of <<extend>> and <<include>> in a UML use-case diagrams. (4marks)
- c) Explain how specialization and generalization relationships are defined. (6marks)
- d) Explain how aggregation and composition relationships are represented in a UML class diagram. (4marks)



J. A.



MULTIMEDIA UNIVERSITY OF KENYA

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

UNIVERSITY EXAMINATIONS 2020/2021

SECOND YEAR FIRST SEMESTER SUPPLEMENTARY/SPECIAL EXAMINATIONS

FOR THE DEGREE OF

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

CCS 2214: OBJECT ORIENTED ANALYSIS AND DESIGN

DATE: 14TH JUNE 2022

TIME: 2 HOURS

INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED.

ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS.

QUESTION ONE (THIRTY MARKS)

a) Explain the following object oriented concepts

- (i) Polymorphism
- (ii) Abstraction
- (iii) Encapsulation
- (iv) Object Oriented Analysis
- (v) Object oriented Design

b) Explain what is meant by "SBI" of an object.

c) Explain the purpose of the following UML diagrams

- i. Use Case diagram
- ii. Sequence diagram
- iii. Component diagram
- iv. Deployment diagram



(5 Marks)
(6 Marks)

v. Class diagram

- d) Explain the various types of classes in an object oriented system (5 Marks)
- e) Describe FOUR software quality factors (6 Marks)
- f) Explain FOUR benefits of object oriented technology (4 Marks)

QUESTION TWO (TWENTY MARKS)

- a) Compare structured and object oriented approach to software development (6 Marks)
- b) Discuss FOUR types of inheritance (8 Marks)
- c) Draw a sample state chart diagram of an object BOOK in MMU library system (6 marks)

QUESTION THREE (TWENTY MARKS)

- a) Define the following types of relationships, giving an example (6 marks)
 - i. Aggregation
 - ii. Composition
 - iii. Generalization
- b) Distinguish between private, protected and public members of a class (6 Marks)
- c) A hockey league is made up of at least four hockey teams. Each hockey team is composed of six to twelve players, and one player captains the team. A team has a name and a record. Players have a number and a position. Hockey teams play games against each other. Each game has a score and a location. Teams are sometimes lead by a coach. A coach has a level of accreditation and a number of years of experience, and can coach multiple teams. Coaches and players are people, and people have names and addresses. Draw a class diagram for this information, and be sure to label all associations with appropriate multiplicities. (8 Marks)

QUESTION FOUR (TWENTY MARKS)

- a) Describe the primary goal of use case modeling (6 Marks)
- b) Distinguish between include and extend relationships between use cases, giving an example of each (8 Marks)
- c) Draw a Sample use case diagram of MMU Library system (6 Marks)

QUESTION FIVE (TWENTY MARKS)

- a) With the aid of a diagram, discuss the phases of RUP. (8 Marks)

b) Describe **THREE** types of software maintenance

(6 Marks)

a) Joe uploads a file into a portal, this portal transfers the file to our server where it is checked for errors. In case of an error the server sends a message to the portal where Joe can see this (if he is still logged in or on his next visit) and upload his file again (hopefully without the errors this time). In case of success the server will also notify the portal but Joe doesn't have to take any action so we are not interested in the result. As the file is okay the system now wait for Sarah who has to start the processing of the file manually after which the process completes. Draw an activity diagram for the system

(6 Marks)





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FACULTY OF COMPUTING & INFORMATION TECHNOLOGY
UNIVERSITY EXAMINATIONS 2019/2020
SECOND YEAR FIRST SEMESTER EXAMINATION FOR THE DEGREE OF
BACHELOR OF COMPUTER SCIENCE/SOFTWARE ENGINEERING

CCS 2212: OBJECT ORIENTED ANALYSIS AND DESIGN

DATE: 6TH DECEMBER 2019

TIME: 2 HOURS

INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED.

ANSWER QUESTION ONE (COMPULSORY) AND ANY OTHER TWO QUESTIONS.

QUESTION ONE (THIRTY MARKS)

- a) Explain the following terms as used in object oriented analysis and Design(OOAD).
- i. Object (2 marks)
 - ii. Abstraction (3 marks)
 - iii. Model (3 marks)
- b) Explain the principle of modularity in the context of modelling using UML. (4 marks)
- c) State the use of the following UML diagrams.
- i. State diagrams (2 marks)
 - ii. Deployment diagrams (2 Marks)
 - iii. Class diagrams (2 Marks)

- d) Using the scenario below, draw a class diagram: A patient will be described by properties patient number, patient name, address which are kept as strings of text and bill kept as currency. A patient can be inpatient or outpatient. Inpatients are admitted and their bill is computed during the date of discharge. outpatients are attended to and their bill calculated during the day of visit. A patient must be attended to by one or two doctors but a doctor may attend zero or more patients in a day. The doctor is described by attributes such as doctor number (string), name (string) and department (string). (12 Marks)

QUESTION TWO (TWENTY MARKS)

- a) Explain the importance of modelling a system. (4 marks)
- b) Identify and explain the building blocks of UML (8 Marks)
- c) Explain the concept of aggregation as used in object oriented analysis and design. Use a diagram to support your answer. (4 marks)
- d) Using an example, describe the principle of association. (4 marks)

QUESTION THREE (TWENTY MARKS)

- a) What is Unified Process Modelling. Explain the iterations, outcomes & Workflow in unified Process Model with neat diagram (8 marks)
- b) Difference between sequence diagram and collaboration diagram (8 marks)
- c) Describe a class diagram, clearly stating where it is used in object oriented analysis and design. (4 marks)

QUESTION FOUR (TWENTY MARKS)

- a) What is the Object Modeling Techniques (OMT). (6 marks)
- b) Explain the importance of Modeling a System. (4 marks)
- c) Compare OOAD and SSAD (Structured Systems Analysis and Design) (6 marks)
- d) Differentiate the concepts of cohesion and coupling. (4 marks)

QUESTION FIVE (TWENTY MARKS)

- a) Draw a Use case diagram for Online Shopping. (8 marks)
- b) Explain the concept of object-oriented design model. (6 marks)
- c) Describe the object-oriented testing strategies (6 marks)



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UNIVERSITY EXAMINATIONS 2019/2020

SECOND YEAR FIRST SEMESTER EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE IN COMPUTER TECHNOLOGY

BCT 2309: OBJECT ORIENTED ANALYSIS AND DESIGN

DATE: 6TH DECEMBER 2019

TIME: 2 HOURS

INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED.

ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS.

QUESTION ONE (THIRTY MARKS)

- Describe five benefits of object oriented system analysis and design over the tradition systems analysis and design methodologies (5marks)
- Explain the importance of UML as a modeling language (5marks)
- In a Use Case Driven Object Oriented Analysis and design; Classes, objects Relationships, Attributes and methods are common terminology used. Explain with the aid of a diagram the meaning of each of these terminologies. (10marks)
- Study the scenario below and create class diagrams for customer and order. Show all the possible attributes and operations.

"We have customers who order our products. We distinguish corporate customers from personal customers, since corporate customers are billed monthly whereas personal customers need to prepay their orders with a credit card."

(6marks)

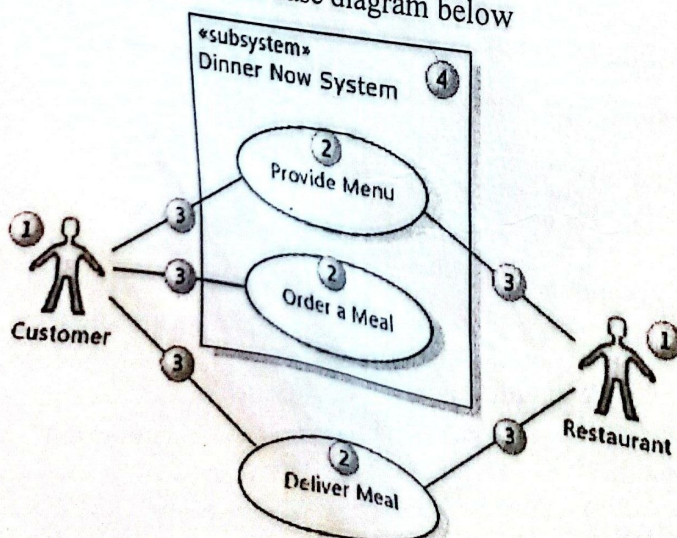
- e) A deployment diagram is made up of 4 components. List and explain each one of these components (4marks)

QUESTION TWO [20 MARKS]

- a) Differentiate between system analyses and system design (4marks)
- b) Describe the following terminologies as used in object oriented analysis and design. (2marks)
- Active object (2marks)
 - Passive objects (2marks)
 - Persistent object (2marks)
 - Non-persistent objects (2marks)
- c) Discuss how a business organization may use the following object oriented relationship terminologies during systems analysis and design. (2marks)
- Association (2marks)
 - Aggregation (2marks)
 - Containment (2marks)
 - Generalization (2marks)

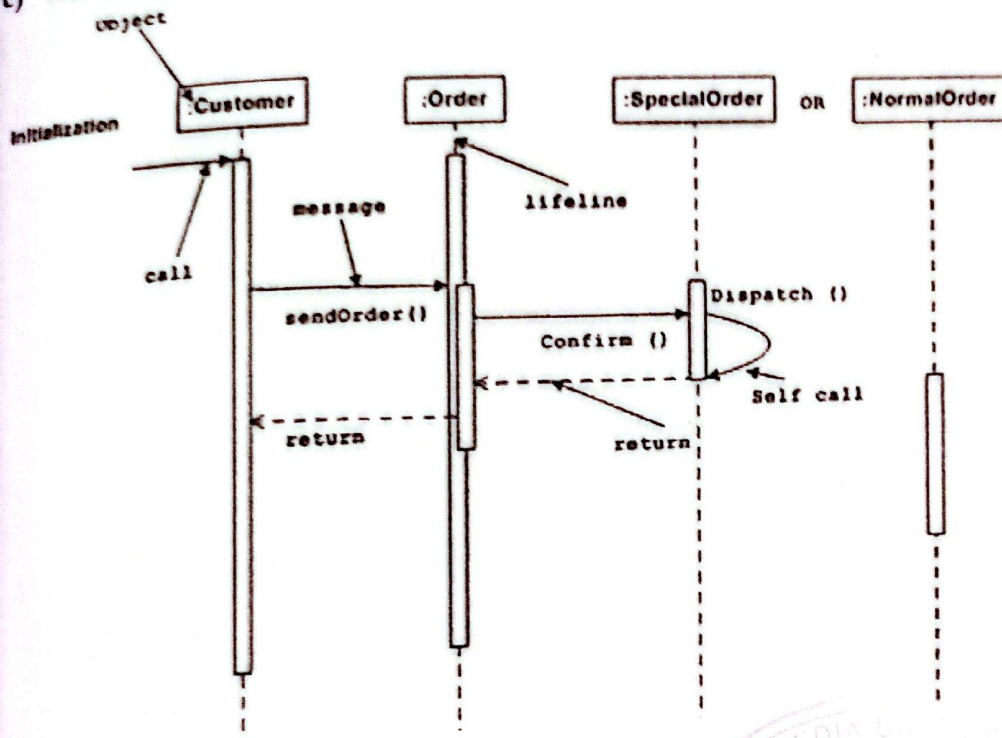
QUESTION THREE [20MARKS]

- a) Outline the process of developing a Use Case Diagram (7marks)
- b) Discuss the pros and con of relying on use case when modeling an object oriented system (6marks)
- c) Interpret the use case diagram below (7marks)



QUESTION FOUR [20MARKS]

- Discuss the roles of activity diagram in object oriented systems analysis (4marks)
- List and describe the various components of a sequence diagram (10marks)
- Generate (narrate) the scenario represented by the sequence diagram below (8marks)



QUESTION FIVE [20MARKS]

- Define activity diagram as used in object oriented systems analysis and design. (2marks)
- Explain the functions of activity diagrams (8marks)
- Using suitable example illustrate how activity diagram is drawn (10marks)



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UNIVERSITY EXAMINATIONS 2020/2021

**SECOND YEAR SECOND SEMESTER EXAMINATIONS FOR THE DEGREE OF
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY/ BACHELOR OF
SCIENCE AND COMPUTER SCIENCE / BACHELOR OF SOFTWARE
ENGINEERING**

BIT 2214 / CCS 2112 : OBJECT ORIENTED SYSTEMS AND DESIGN

DATE: 6TH DECEMBER 2021

TIME: 2 HOURS

INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED.

ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS.

QUESTION ONE (THIRTY MARKS)

- a) State three goals of Object Oriented analysis. **(3 Marks)**
- b) State any five characteristics of objects as used in Object-Oriented Systems and Design. **(5 Marks)**
- c) The problem domain is an information system for a video rental store. Simplifying assumptions and details:
- It is a stand-alone store, not part of a larger organization.
 - Rents only videos, not computer games or other items.

- A “video” can be in any medium: tape, DVD, and so on.
 - The rental charge may vary by medium. For example, DVD rentals are more expensive than tapes.
 - The store does not sell anything. For example, there are no sales of videos or food.
 - All transactions are rentals.
 - The input medium by which membership and video rentals are captured is not important.
 - Cash-only payments.
 - On completion of a rental, the customer receives a transaction report with ‘typical’ information on—use your judgement.
 - Each renter has a separate membership.
- i. Identify at least 4 primary use cases and related actors. Draw the results in a UML use case diagram. (5 Marks)
 - ii. Write the *Rent Videos* use case in expanded, essential, conversational style. Include both a *typical course of events*, and an *alternatives* section. In the alternatives section, list exactly two frequent alternatives. Do **not** write an elaborate heading section for the use case (i.e., ignore Purpose, Overview, ...); rather, start immediately with the typical course of events. (5 Marks)
 - iii. Bounded by the provided *Rent Videos* use case, create a conceptual model and illustrate it in UML notation. Show concepts, associations, and attributes. (8 Marks)
 - iv. Bounded by the provided *Rent Videos* use case, create a system sequence diagram in the UML notation. Show actors and system events. (4 Marks)

QUESTION TWO [20 MARKS]

- a) Define UML. (2 Marks)
- b) State and briefly explain the two relationships in UML, giving their properties. (6 Marks)
- c) In the approach to “analysis” and “design” that has been studied in this course, *briefly* compare and contrast these two terms. What is “analysis” about? What is “design” about? (4 Marks)

- d) Briefly, what are some motivations for creating 1) conceptual models, and 2) operation contracts? (4 Marks)
- e) Explain the difference between aggregation and composition with examples. (4marks)

QUESTION THREE [20 MARKS]

- a) Draw a simple Use Case diagram showing all first main functions of the ATM system. (8 Marks)
- b) Draw an activity diagram for Withdraw Cash transaction. (12 Marks)

QUESTION FOUR [20 MARKS]

- a) Explain the following concepts relating them with UML: (8 Marks)
- Forward engineering (Proactive approach)
 - Reverse engineering (Reactive approach)
 - Coupling
 - Cohesion
- b) Consider the world of libraries. A library has books, videos, and CDs that it loans to its users. All library material has a id# and a title. In addition, books have one or more authors, videos have one producer and one or more actors, while CDs have one or more entertainers. The library maintains one or more copies of each library item (book, video or CD).
- Copies of all library material can be loaned to users. Reference-only material is loaned for 2hrs and can't be removed from the library. Other material can be loaned for 2 weeks. For every loan, the library records the user, the loan date and time, and the return date and time. For users, the library maintains their name, address and phone number. Draw a class diagram (or two, if this is more convenient) for the description above. Make sure to show attributes, multiplicities and aggregations/compositions, where appropriate. (12 Marks)
- No need to show any operations.

QUESTION FIVE [20 MARKS]

- a) Give a state diagram that describes the process of passing a graduate course as a set of concurrent activities. The process has as follows: To pass, a student has to attend all but two lectures, present to the class a paper she read, and complete a course project, due on the last day of the term. To give her presentation, the student is given a date by the

instructor, prepares her presentation, and gives it on the assigned day. At any time, the student can drop the course. Make sure to define events, conditions, actions for transitions in your diagram, where appropriate.

(10 Marks)

- b) To give an exam, an instructor first notifies the students of the exam date and the material to be covered. She then prepares the exam paper (with sample solutions), gets it copied to produce enough copies for the class, and hands it out to students on the designated time and location. The students write their answers to exam questions and hand in their papers to the instructor. The instructor then gives the exam papers to the TAs, along with sample solutions to each question, and gets them to mark it. She then records all marks and returns the papers to the students.

Draw a sequence diagram that represents this process. Make sure to show when is each actor participating in the process. Also, show the operation that is carried out during each interaction, and what its arguments are.

(10 Marks)